PART 1 · STUD WALL · HORNBACH CUT SHEET

Stud wall plan

Insulated stud wall in front of the gips back wall. The frame holds itself up and is fixed into the floor, the ceiling joists, the brick pillar and the brick kitchen wall. A 700 × 1180 bay by the kitchen wall stays open, so the existing tap, sockets and pipes stay where they are. Hornbach cuts every piece to final size. All sizes are in millimetres.

Wall 2605 wide, ceiling 2450–2473 | Open bay 700 × 1180 | Frame 44 × 94 vuren | OSB/3 18 mm | 10 bars of 300 cm | 3 OSB sheets | Finish painted OSB

THE WALL TODAY

Starting point



Seen from inside the bijkeuken

Left: the exterior wall (gips) with the vent grille. Middle: the brick pillar. Right: the brick kitchen wall and window. The tap, drain, socket and junction box at the bottom right all stay reachable in the open bay. The ceiling slopes from 2450 at the exterior wall to 2473 at the kitchen wall.

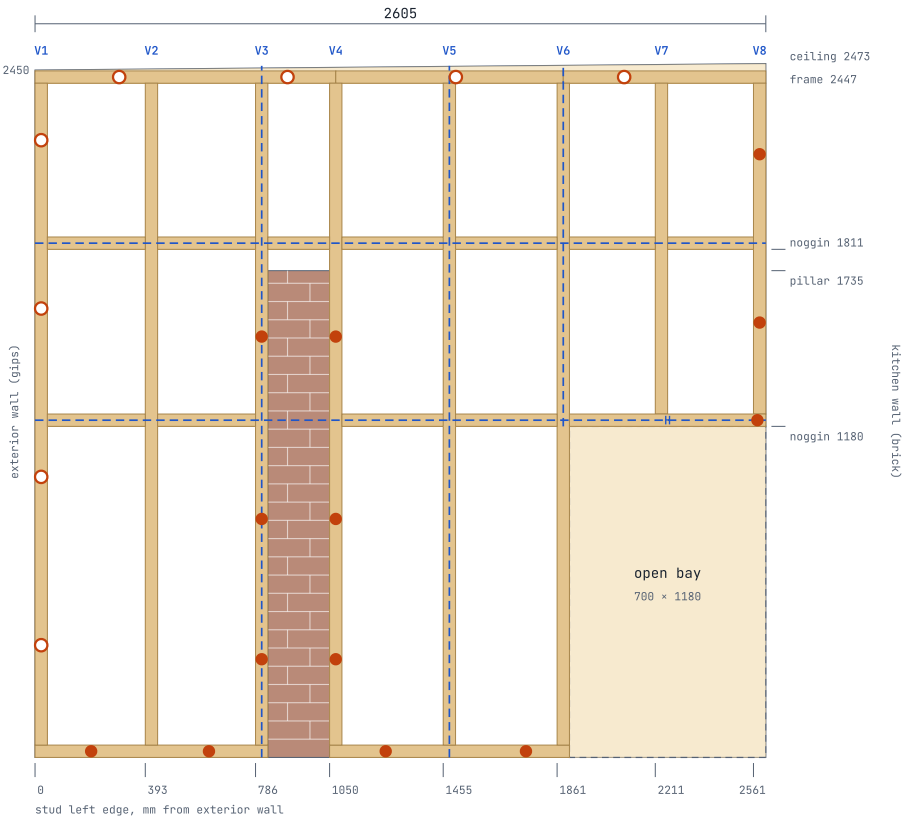
Check the vent grille

The frame stands 35 mm off the back wall to clear the water pipe, so the new wall sticks out 147 mm. Measure from the back wall to the nearest edge of the grille at the top left. If it's less than about 150 mm, the wall would cover part of it. Tell me before you build that corner.

ELEVATION · SEEN FROM INSIDE THE BIJKEUKEN

Frame layout

The first noggins have their bottom face at 1180, and the second row at 1811. The open bay by the kitchen wall has no floor plate (pipes) and no stud at the bottom. V6 frames its left side. A header at 1180 spans over it, and the short studs V7 and V8 stand on that header. The frame is built level with its top at 2447. The thin band above it is the gap to the sloping ceiling, filled with packers. The blue dashed lines are the OSB joints. Drawn to scale: 1 drawing unit = 1 mm.

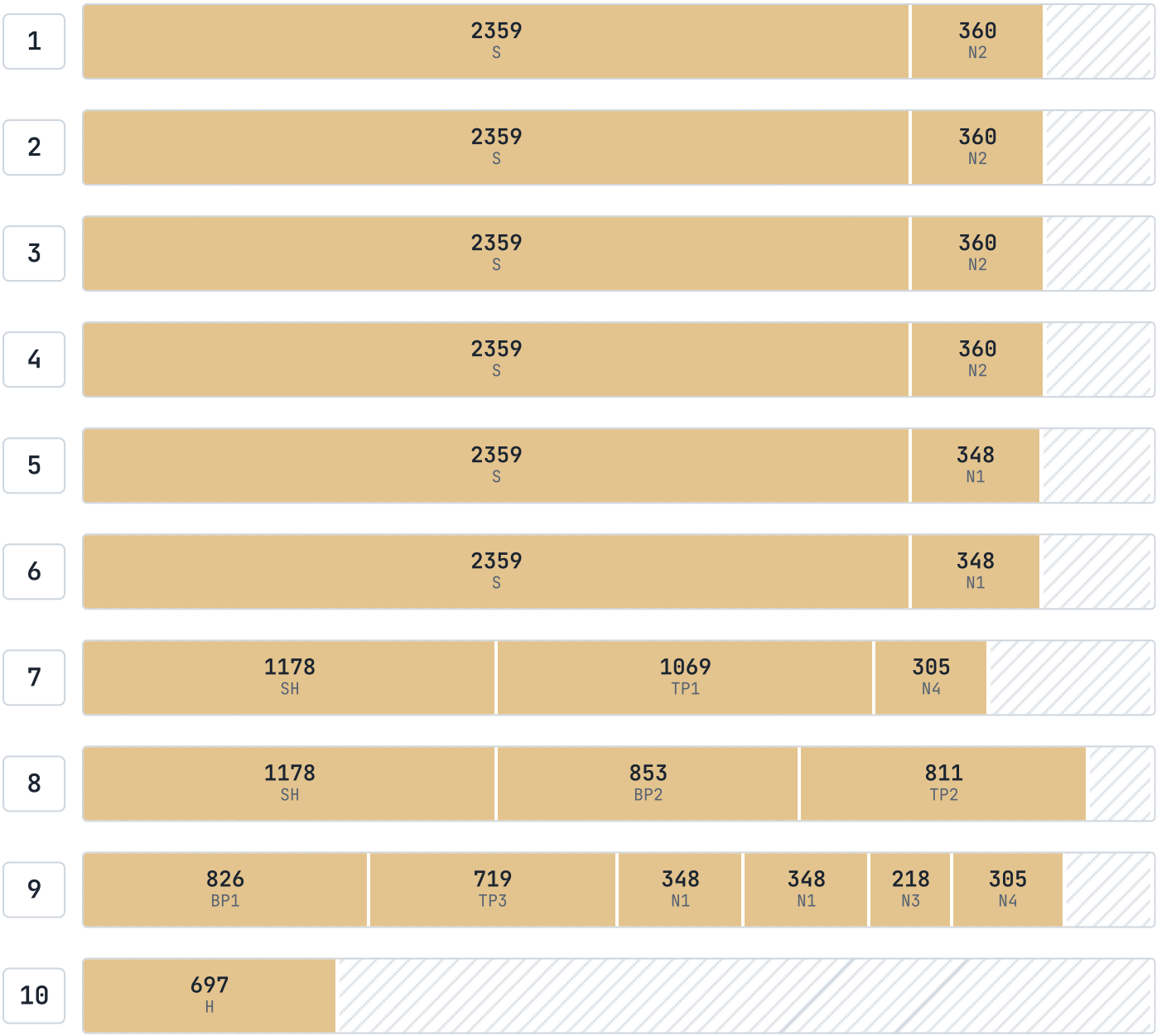


44×94 timber · Brick pillar (74 deep, OSB 55 mm in front) · Plug into brick or floor · Screw into a joist or batten behind the gips · Open bay, no OSB

TIMBER · 44 × 94 VUREN · 10 BARS OF 300 CM

Bar plan for the zaagservice

Show this to the saw operator. Tap a bar once it's cut. The hatched part is offcut. Write the code on each piece with a marker. Bar 10 is mostly spare, for blocking and mistakes.


1
2359
S
360
N2
2
2359
S
360
N2
3
2359
S
360
N2
4
2359
S
360
N2
5
2359
S
348
N1
6
2359
S
348
N1
7
1178
SH
1069
TP1
305
N4
8
1178
SH
853
BP2
811
TP2
9
826
BP1
719
TP3
348
N1
348
N1
218
N3
305
N4
10
697
H

CODE	LENGTH	QTY	WHERE IT GOES
S	2359	6	Full studs V1–V6, floor plate → top plate
SH	1178	2	Short studs V7 and V8, header → top plate
H	697	1	Header over the open bay, bottom face at 1180, V6 → kitchen wall
TP1	1069	1	Top plate, exterior wall → centre of V4
TP2	811	1	Top plate, V4 centre → V6 centre
TP3	719	1	Top plate, V6 centre → kitchen wall
BP1	826	1	Floor plate, exterior wall → pillar
BP2	853	1	Floor plate, pillar → open bay (runs under V6)
N1	348	4	Noggins V1–V2 and V2–V3, both rows
N2	360	4	Noggins V4–V5 and V5–V6, both rows
N3	218	1	Noggin above the pillar, V3–V4, row 2
N4	305	2	Noggins V6–V7 and V7–V8, row 2

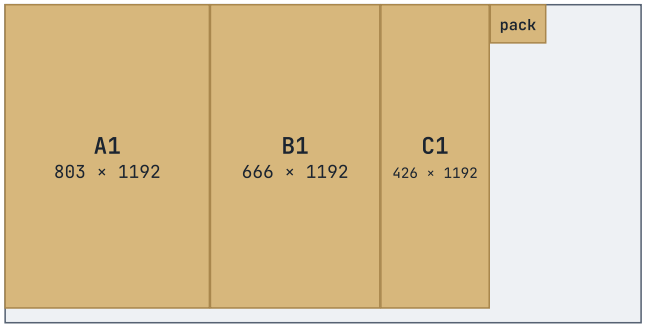
OSB/3 · 18 MM · 3 SHEETS OF 2500 × 1250

Sheet plan

Sheet 1 gives the three tall bottom panels. Sheets 2 and 3 are first cut into strips across the long side, then each strip is cut to width. Panel codes match the wall layout below: column A–D, row 1 at the bottom. Sizes are wall width × wall height. Tap a sheet once it's cut.

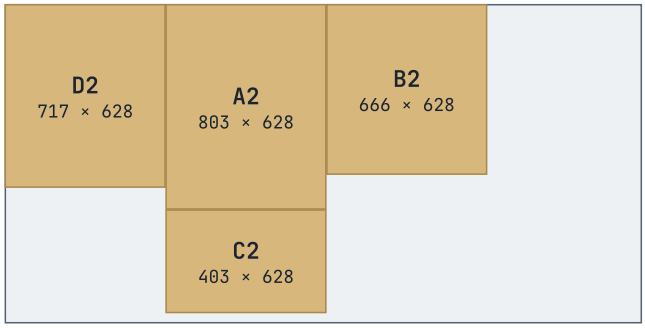
Sheet 1

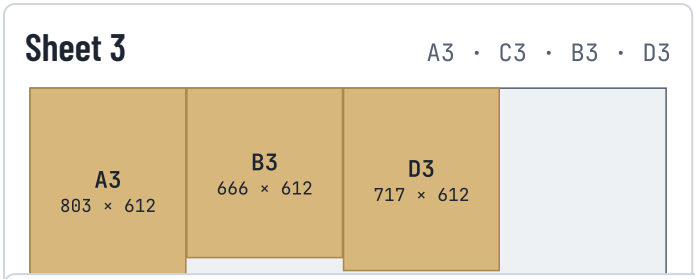
A1 · B1 · C1 + packing

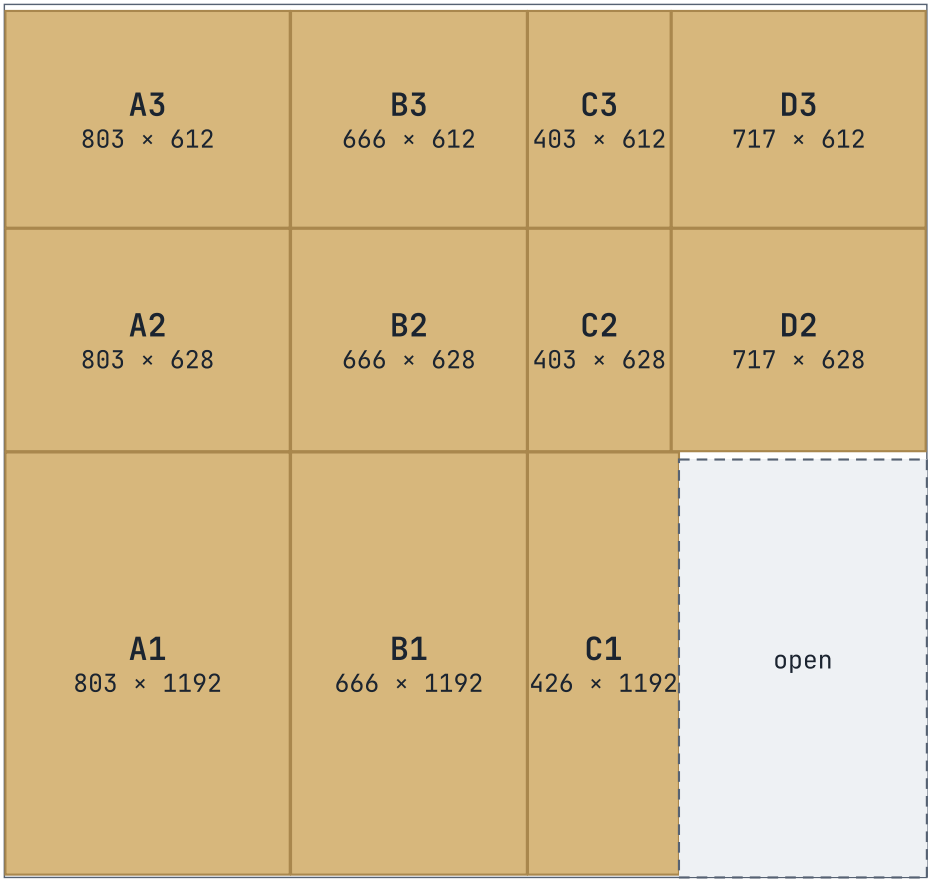


Sheet 2

D2 · A2 · C2 · B2







Gaps: 8 mm at the floor, 3 mm at joints and walls. Row 3 ends flush with the top of the frame at 2447. The gap above it (3 mm at the exterior wall, 26 mm at the kitchen wall) gets a ceiling trim strip at the end. C1 ends flush with the inside face of V6, so the left edge of the open bay is clean. D2 sits in row 2 like the other panels, so the bottom 24 mm of the header shows above the bay. Paint it or cover it with a lat.

SHOPPING LIST · TRIP 1

Everything else for the frame and OSB

10× vuren 44×94, 300 cm
3× OSB/3 18 mm, 2500 × 1250
~5 m² glaswol plaat, ~100 mm

It squeezes slightly into the 94 mm cavity. The open bay stays empty.

60× hoekanker (90° angle bracket) 70×70×55, galvanised

49 needed: 19 for the stud ends, 8 for the open bay, 22 for the noggins

500× ankerschroef 5×35

8 per bracket

20× kozijnplug 10×100

Floor and V8 into the kitchen wall (a few spare)

16× plug 6 mm + screw 5×50

Header bracket, packing piece, and the small pillar brackets

6× small angle bracket, legs 50 mm or less (e.g. 50×50×35)

Holds V3 and V4 to the pillar's front face. A 70 leg won't fit: only 55 mm of stud sticks out past the pillar.

~10× spring toggles (hollow-wall anchors)

Backup for the gips if a joist or batten can't be found

50× constructieschroef 6×120

V1 into the battens, header, spare

~20× constructieschroef 6×160

Top plate through the packers into the joists, and studs into battens behind the gips

300× spaanplaatschroef 4×45

OSB: every 150 mm along edges, every 300 mm in the field

Detector that finds wood, metal and live cables

For the ceiling joists and the battens behind the gips, and to avoid cables before drilling

Packers (uitvulplaatjes), mixed 2–10 mm

To fill the 3–26 mm gap above the top plate at every joist, and to level the floor plate

10 mm stone drill bit, ≥ 150 mm long

Marker, mask (FFP2), gloves

FINISH · AFTER THE OSB IS UP

Paint the OSB

The OSB is the finished wall: no extra layer on top. The cupboard, worktop and wall strips screw straight into it anywhere. Where it stays visible, it gets an insulating primer and two coats of washable satin paint. The chip texture stays visible under the paint.

Shopping list · finish

Insulating primer (isolerende / vlekkenblokkerende primer), ~2.5 L

Stops the marker codes and the resin in the OSB from bleeding through the white paint

Washable satin wall paint (zijdeglans muurverf), white, ~2.5 L

Two coats. Satin wipes clean better than matte in a laundry room.

Roller (short pile) + tray, brush for the edges, masking tape

2× acrylic caulk (acrylaatkit, overschilderbaar) + caulk gun

Joints with the side walls and the ceiling

1× ceiling trim strip (plafondlijst), at least 40 mm, 2.6 m + mounting kit

Covers the 3–26 mm gap between the wall and the sloping ceiling

FIXING

Where the frame is held

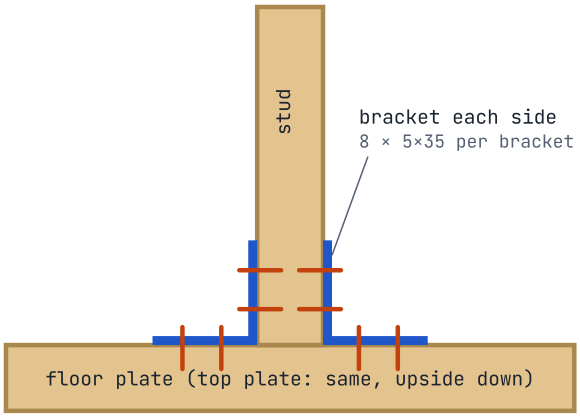
The floor, the ceiling joists, the pillar and the kitchen wall carry the frame. Timber-to-timber joints use 90° angle brackets (hoekankers). The OSB screwed over everything makes it one stiff panel.

WHERE	PIECE	FIXING
Studs ↔ plates	V1–V8	Two angle brackets per stud end, one on each side face. Against the pillar or kitchen wall only the free side gets one.
Noggins ↔ studs	N1–N4	One angle bracket under each end. Screw the bracket to the stud at the right height first, then drop the noggin on it.
Floor	BP1, BP2	Plug 10×100 through the plate every ~50 cm, between the studs. BP2 stops at the open bay, clear of the pipes.
Ceiling joists	TP1–3	The ceiling slopes, so the gap above the level plate grows from ~3 mm (exterior wall) to ~26 mm (kitchen wall). Put stacked packers at every joist, then 6×160 up through plate and packers into the joist. If the joists run parallel to the wall and the plate misses them: spring toggles through the gips every ~40 cm.
Pillar sides (brick)	V3, V4	3 small angle brackets each (legs ≤ 50 mm) in the front corner: one leg on the stud side, one on the pillar front face with a 6 mm plug + 5×50. No plugs through the stud: with the 35 mm gap the overlap is only 39 mm.
Kitchen wall (brick)	V8, H	V8: 2 × plug 10×100. Header: an angle bracket on the brick with 6 mm plugs and 5×50 screws.
Exterior wall (gips)	V1	6×120 into the battens behind the gips (detector). No batten found: 4 spring toggles.

DETAILS · 1 DRAWING UNIT = 1 MM

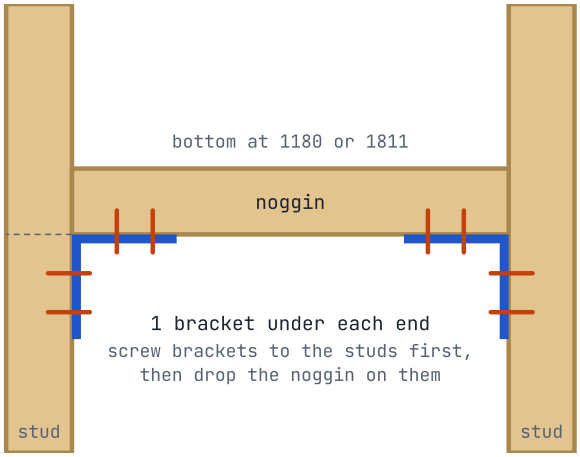
How the pieces connect

Every timber joint uses a 90° angle bracket (hoekanker 70×70×55, 2.5 mm) with 5×35 anchor screws, 4 per leg. Brackets always go on the **side faces** of the timber, inside the 94 mm depth. Never put one on the front face, where the OSB goes, or on the back face, against the wall.



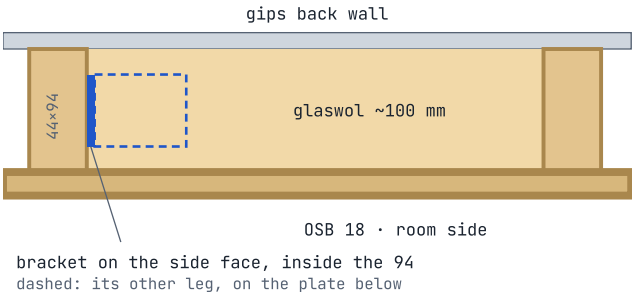
Stud to plate

Stand the stud on its mark and check it's plumb. Screw one bracket to each side face, at the bottom and the top. V7 and V8 sit on the header the same way.



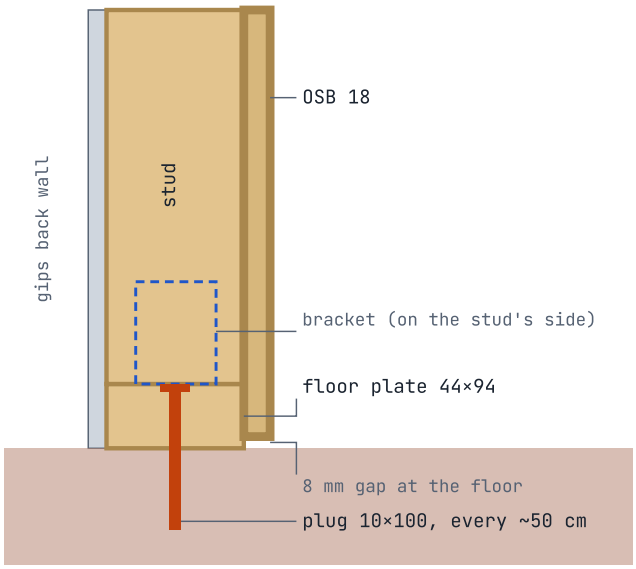
Noggin to studs

The brackets hold the noggin up like small shelves, so you don't have to hold the noggin while you screw. The header H over the open bay goes in the same way: a bracket on V6, and a bracket plugged into the kitchen wall brick.



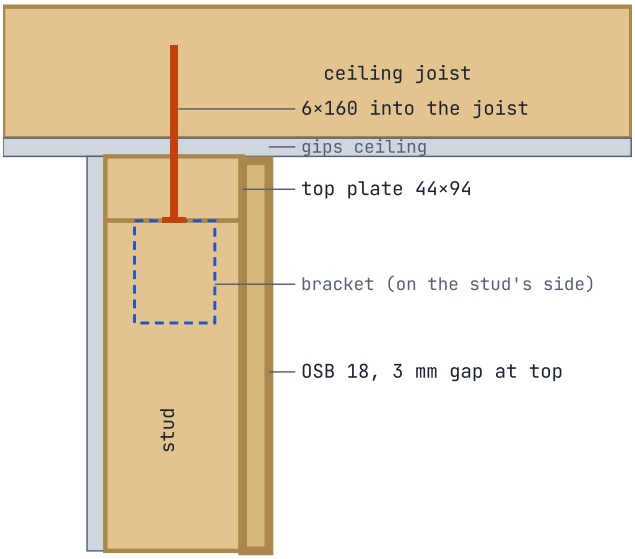
Seen from above

The stud's 94 side runs away from the wall. Bracket legs are 55 wide, so they fit within the 94 and never touch the gips or the OSB.



Floor plate

35 mm out from the gips, with the water pipe running free behind it. Drill through the plate into the floor and hammer in the plugs, between the stud positions so they don't clash with the brackets.



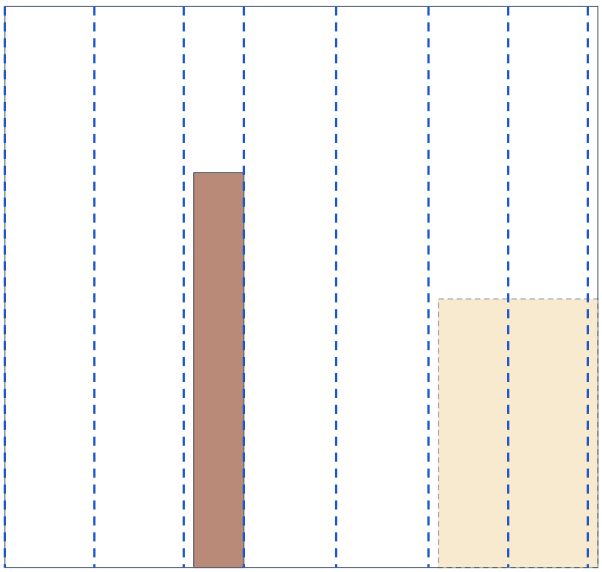
Top plate

Hold it right above the floor plate (check with the waterpas). The ceiling slopes, so the gap above the plate grows from about 3 mm at the exterior wall to 26 mm at the kitchen wall. Fill it with stacked packers at every joist, then screw through plate and packers into the joist. The plate stays level and straight.

BUILD JOURNEY

Step by step

The blue parts are the ones added in each step. Lay out the pieces for a step, check them against the parts list, then build.

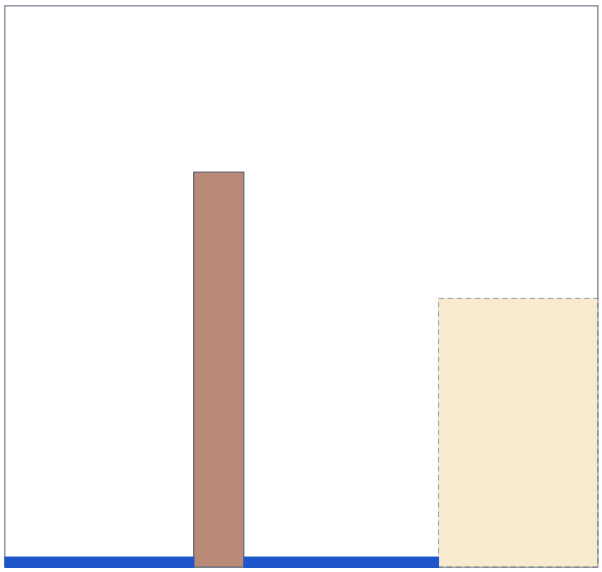


1 Mark out

Scan the ceiling, the exterior wall and the back wall with the detector, and mark joists, battens and cables. Mark the stud left edges on the floor and the ceiling.

`detector` `waterpas` `pencil`

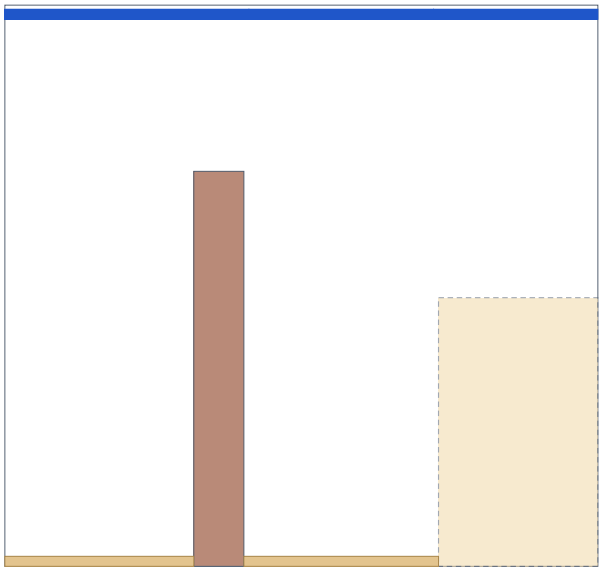
`0 · 393 · 786 · 1050 · 1455 · 1861 · 2211 · 2561`



2 Floor plate

Check the floor with the waterpas and shim under the plates if it slopes, so the frame starts level. Lay BP1 (exterior wall → pillar) and BP2 (pillar → 1905, where the open bay starts) 35 mm out from the back wall, so the water pipe runs free behind them. Drill through the plates into the floor and plug them every ~50 cm, between the stud marks.

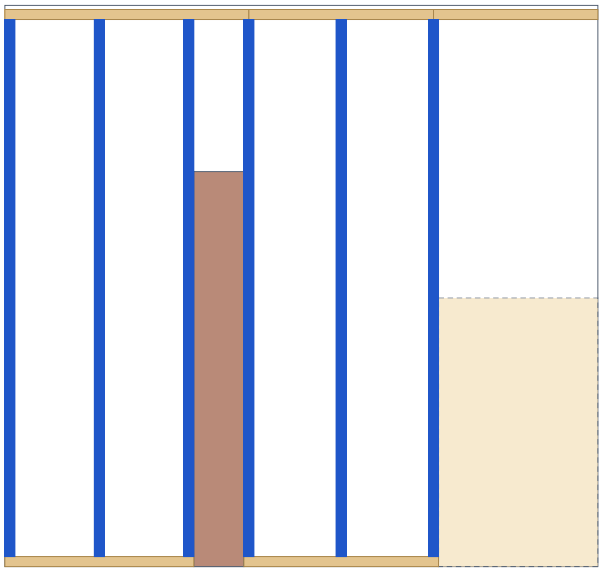
`1×` BP1 826 `1×` BP2 853 `5×` plug 10×100



3 Top plate

Keep the top plate level, not following the ceiling. Its top sits at 2447, straight above the floor plate. The gap to the ceiling grows from ~3 mm at the exterior wall to ~26 mm at the kitchen wall. Stack packers in it at every joist and screw through them into the joist.

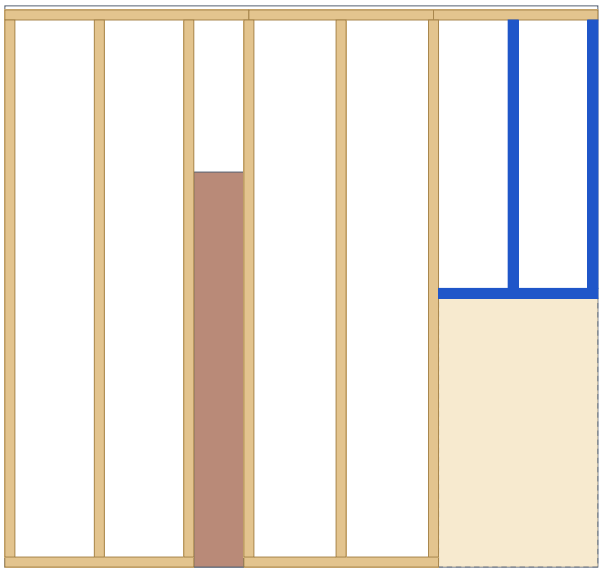
`1×` TP1 1069 `1×` TP2 811 `1×` TP3 719 `~8×` 6×160 packers up to 26 mm



4 Studs V1–V6

Stand each stud on its mark, check it's plumb, and fix it with a bracket on each side at the top and bottom. V3 and V4 go tight against the pillar: brackets on the free side only, plus 3 small brackets each from the stud side to the pillar front face (at 300, 750 and 1500). Screw V1 into the battens behind the gips.

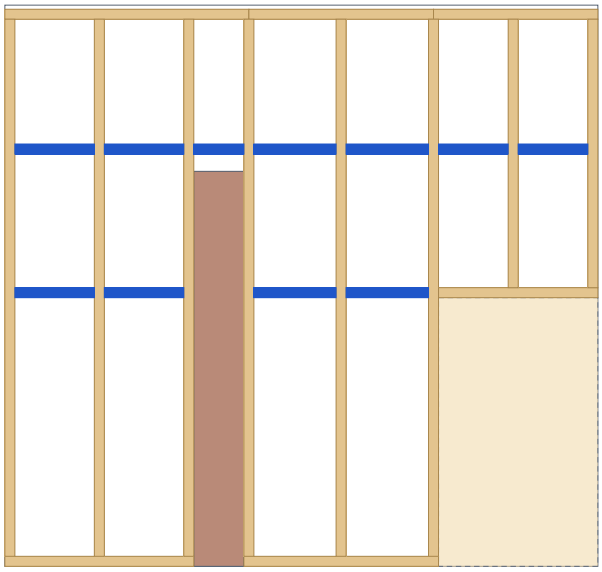
`6×` S 2359 `19×` bracket `6×` small bracket `6×` plug 6 mm + 5×50 `~4×` 6×120



5 Open bay

Screw a bracket to V6 with its top at 1180, and one to the kitchen-wall brick. Drop the header H onto them and screw up into it. Stand V7 and V8 on the header, with brackets at the top and bottom. Plug V8 into the kitchen wall.

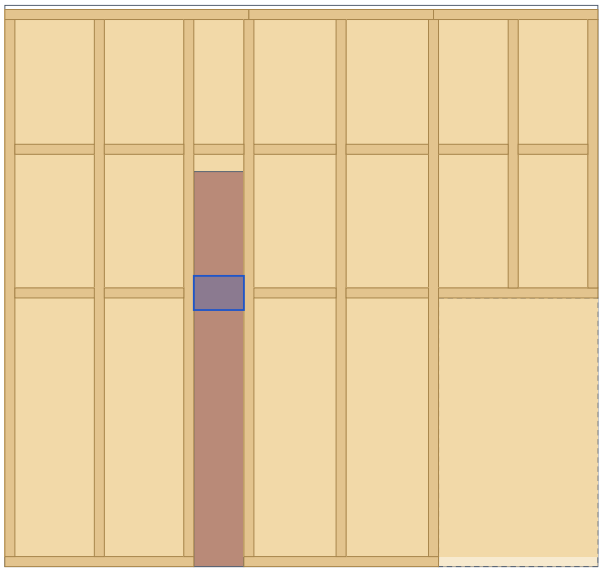
`1× H 697` `2× SH 1178` `8× bracket`
`2× plug 10×100` `4× plug 6 mm + 5×50`



6 Noggins

In every bay, screw a bracket to both studs with its top at 1180 (row 1) and 1811 (row 2). Drop the noggin on and screw up into it. Row 1 skips the pillar bay and the open bay. N3 goes above the pillar in row 2.

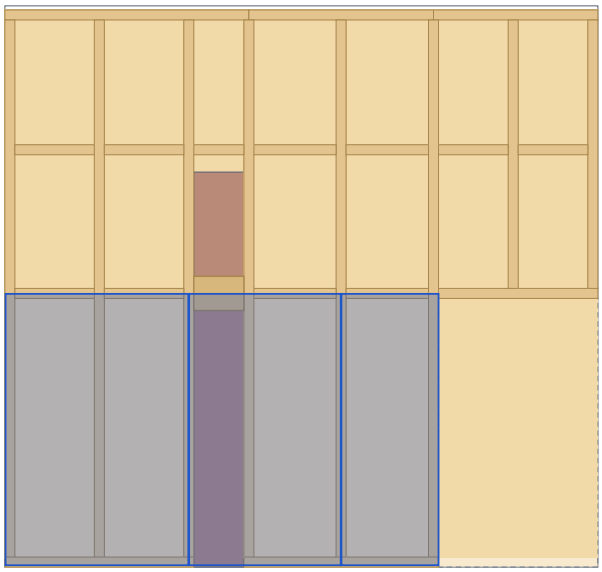
`4× N1 348` `4× N2 360` `1× N3 218`
`2× N4 305` `22× bracket`



7 Insulate

Finish any new pipe or socket work first. Press glass wool into every closed bay, snug but not crushed. Screw a 218 × 150 packing block to the pillar face (a 44×94 offcut laid flat plus ~11 mm of packers, to fill the 55 mm gap), centred at 1202, so the OSB joint has something behind it.

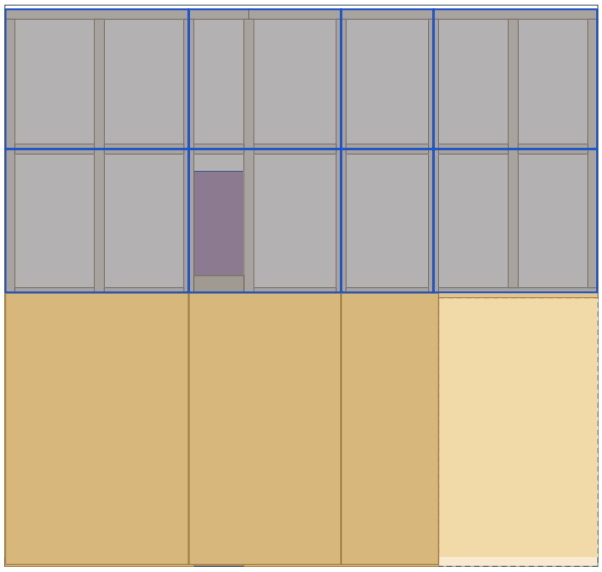
`~5 m²  glaswol` `1× packing 218×150`
`mask + gloves`



8 OSB row 1

Stand A1, B1 and C1 on 8 mm spacers, with 3 mm gaps between them. C1 ends flush with V6 at the open bay. Screw every 150 mm along the edges and every 300 mm in the middle.

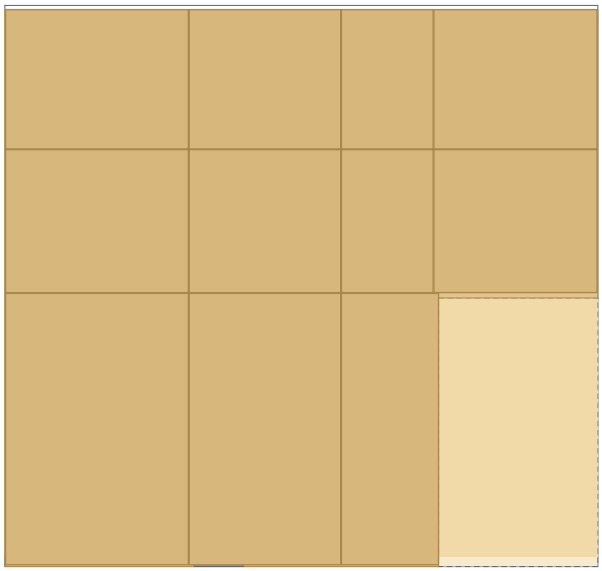
`1×` A1 803×1192 `1×` B1 666×1192 `1×` C1 426×1192 `~90×` 4×45



9 OSB rows 2 and 3

Row 2 first (A2, B2, C2, D2; D2 sits on the header), then row 3, ending flush with the top of the frame. The frame and OSB are done.

`4×` row 2 panels `4×` row 3 panels `~200×` 4×45



10 Paint & finish

Caulk the joints with the side walls, and fix the ceiling trim over the gap at the top. Sand off any splinters, then prime the OSB with the insulating primer so the marker codes and resin stains don't bleed through. Finish with two coats of washable satin paint, including the bottom of the header and the edges around the open bay.

caulk ceiling trim insulating primer satin paint